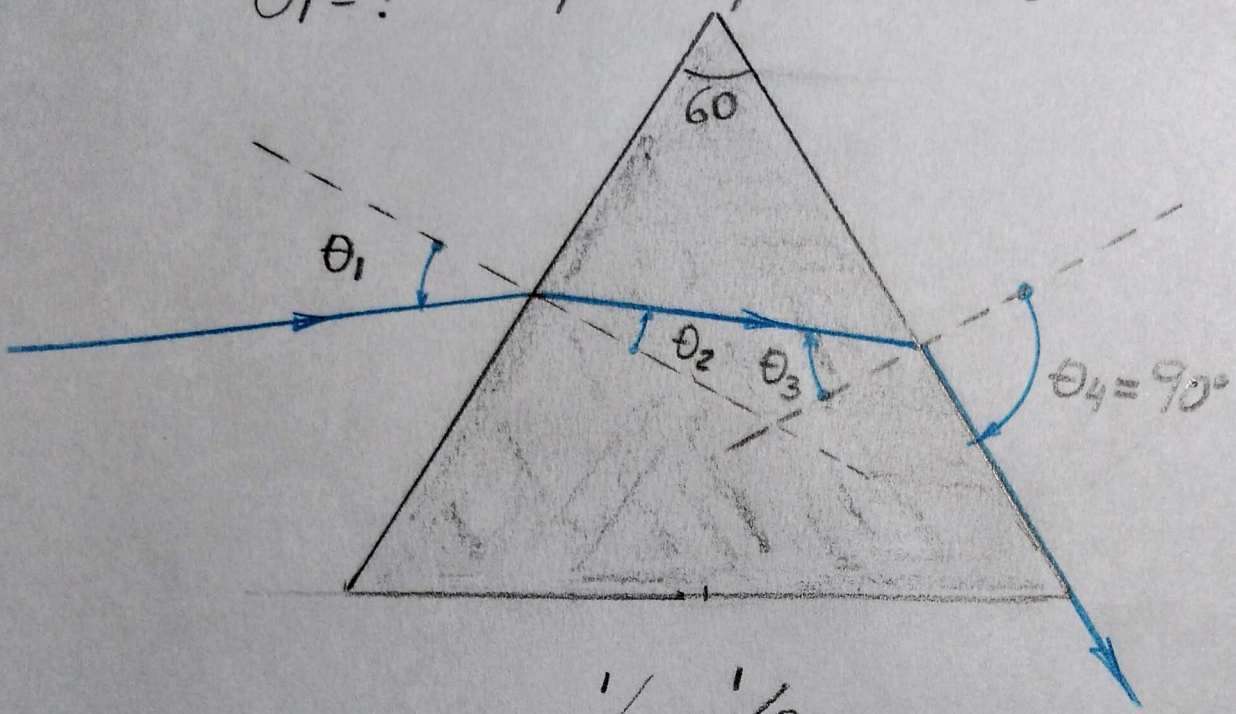
30) a) $\phi = 60°$ $n_v = 1,5$ $\theta_1 = ?$ →

$\theta_1 = ?$ → para que emerja en 2ª interfase



$n_v \operatorname{sen}\theta_3 = n_{aire} \operatorname{sen}\theta_4$

$\theta_3 = \text{arc sen}\left(\frac{1}{1,5}\right) = 41,8°$

$\theta_2 = \phi - \theta_3 = 60 - 41,8 = 18,2°$

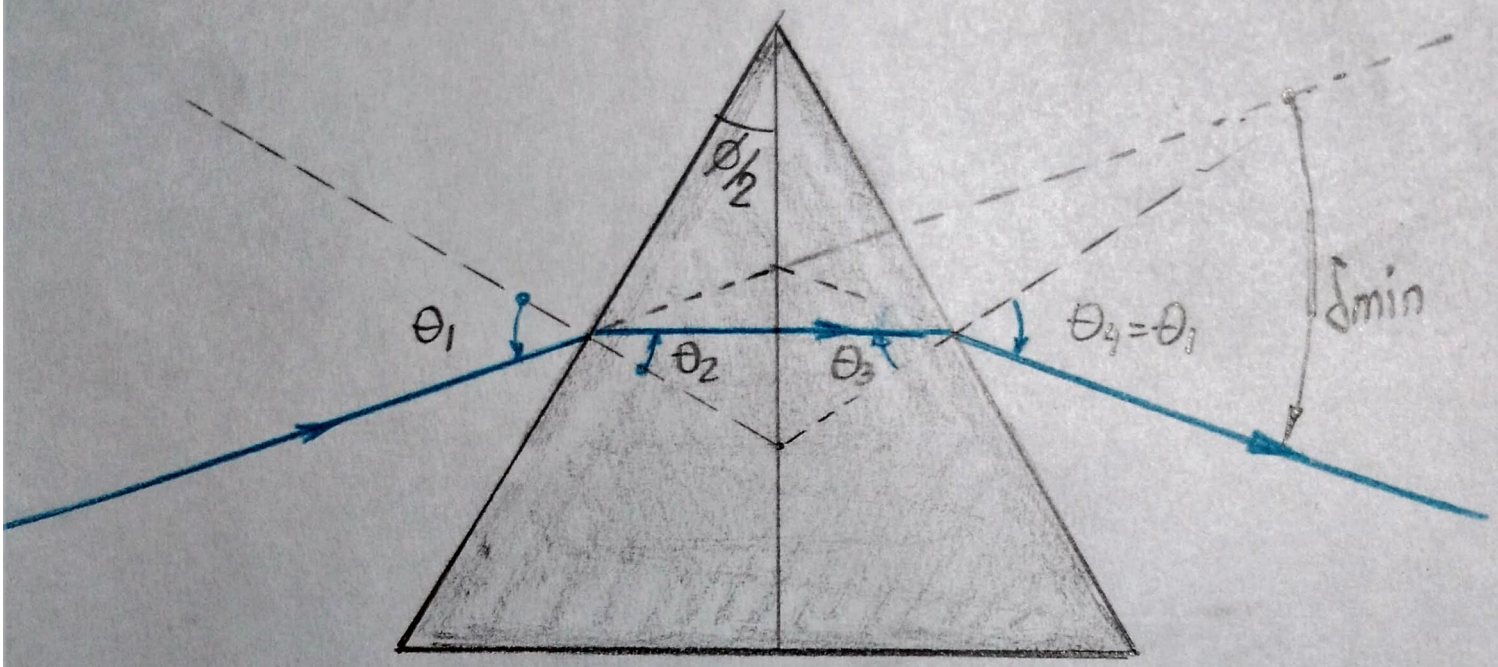
$n_{aire} \operatorname{sen}\theta_1 = n_v \operatorname{sen}\theta_2$

$\theta_1 = \text{arcsen}\left(1,5 \cdot \operatorname{sen} 18,2\right) = 27,9°$

para que emerja $\theta_1 > 27,9°$

30) b) $\phi = 60°$ $n_v = 1{,}5$

$\theta_1 = ?$ ⟶ $\theta_1 = \theta_4$



$$\theta_2 = \frac{\phi}{2} = \frac{60}{2} = 30°$$

$n_{ar} \operatorname{sen}\theta_1 = n_v \operatorname{sen}\theta_2$

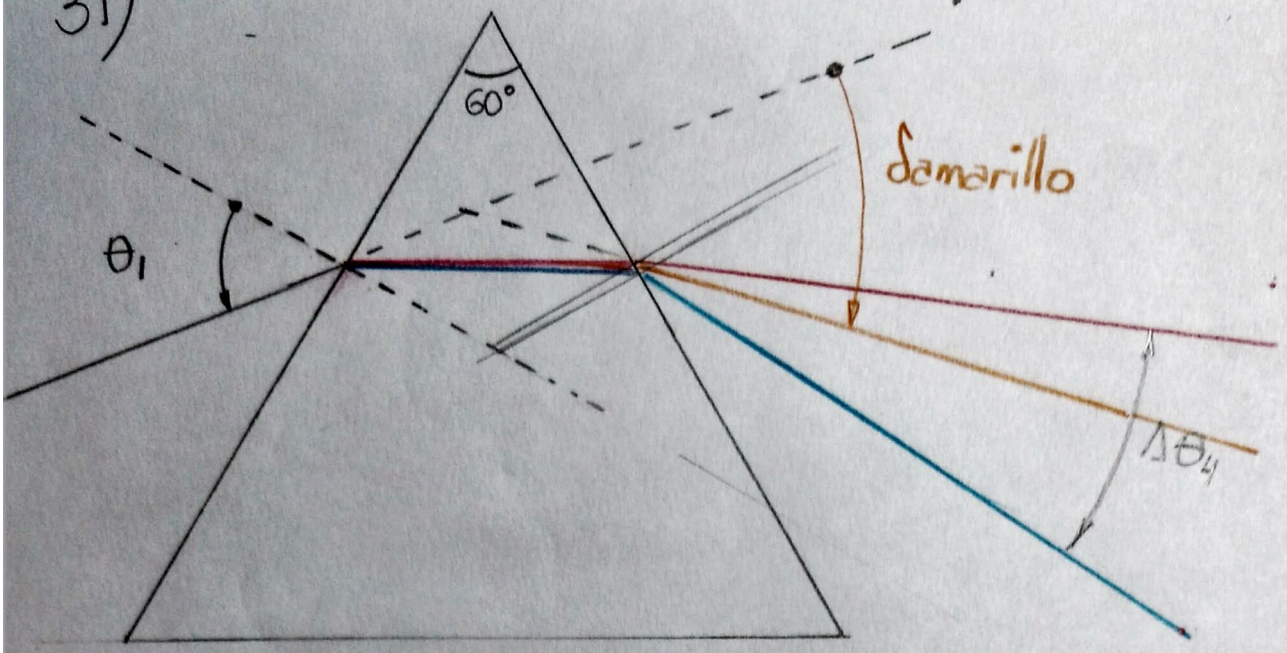
$$\theta_1 = \operatorname{arcsen}(1{,}5 \operatorname{sen} 30) = 48{,}6°$$

$$\delta_{min} = \theta_1 - \theta_2 + \theta_4 - \theta_3 = 48{,}6 - 30 + 48{,}6 - 30$$

$$\delta_{min} = 37{,}2°$$

31)

$n_V = 1{,}66$ $\quad n_R = 1{,}62$



Violeta:

$$\cancel{n_{aire}}\ \text{sen}\,\theta_1 = n_V \,\text{sen}\,\theta_{2V}$$

$$\theta_{2V} = \text{arc sen}\,\frac{\text{sen}\,50^\circ}{1{,}66} = 27{,}5^\circ$$

$$\theta_{3V} = \phi - \theta_{2V} = 60^\circ - 27{,}5^\circ = 32{,}5^\circ$$

$$n_V\ \text{sen}\,\theta_{3V} = \cancel{n_{aire}}\ \text{sen}\,\theta_{4V}$$

$$\theta_{4V} = \text{arc sen}\,(1{,}66 \cdot \text{sen}\,32{,}5) = 63{,}1^\circ$$

Rojo:

$$\cancel{n_{aire}}\ \text{sen}\,\theta_1 = n_R \,\text{sen}\,\theta_{2R}$$

$$\theta_{2R} = \text{arc sen}\,\frac{\text{sen}\,50}{1{,}62} = 28{,}2^\circ$$

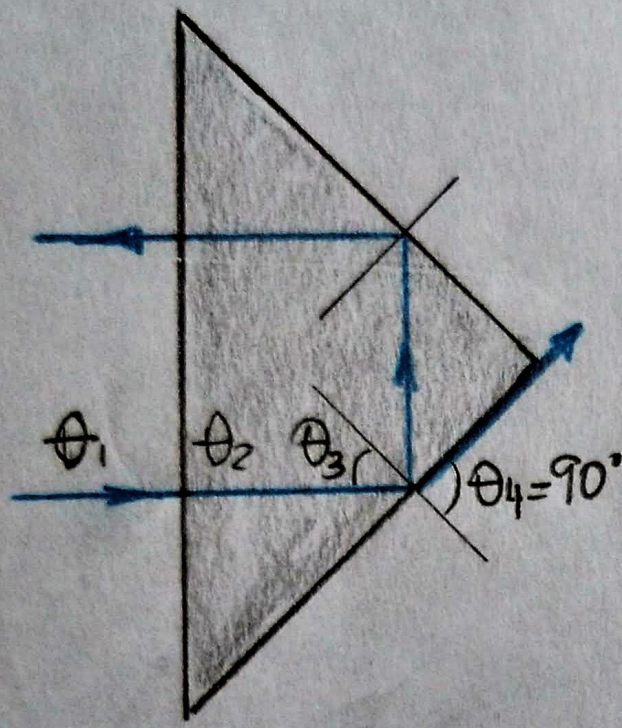
$$\theta_{3R} = \phi - \theta_{2R} = 60^\circ - 28{,}2^\circ = 31{,}8$$

$$n_R\ \text{sen}_{3R} = \cancel{n_{aire}}\ \text{sen}\,\theta_{4R}$$

$$\theta_{4R} = \text{arc sen}\,(1{,}62 \cdot \text{sen}\,31{,}8^\circ) = 58{,}6^\circ$$

$$\text{dispersión angular} = \theta_{4V} - \theta_{4R} = 63{,}1 - 58{,}6 = 4{,}5^\circ$$

32)



$\theta_1 = \theta_2 = 0$

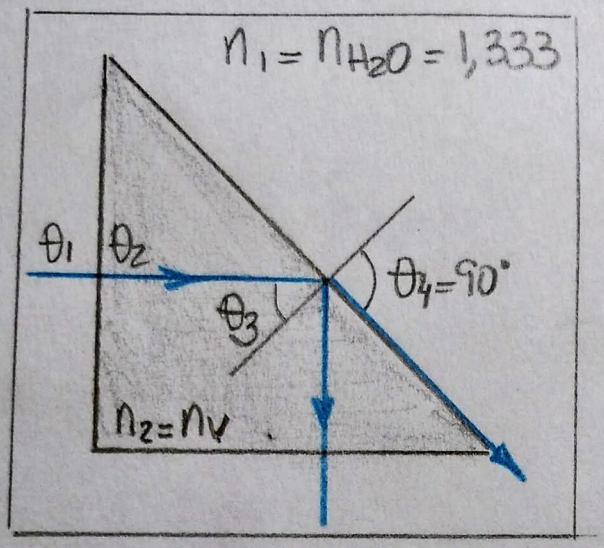
$n_v \operatorname{sen}\theta_3 = n_{ar} \operatorname{sen}\theta_4$

$\theta_3 = 45°$ $\quad$ $\theta_4 = 90$

$$n_v = \frac{1}{\operatorname{sen} 45°} = 1{,}414$$

$n_v > 1{,}414$

33)



$\theta_1 = \theta_2 = 0$

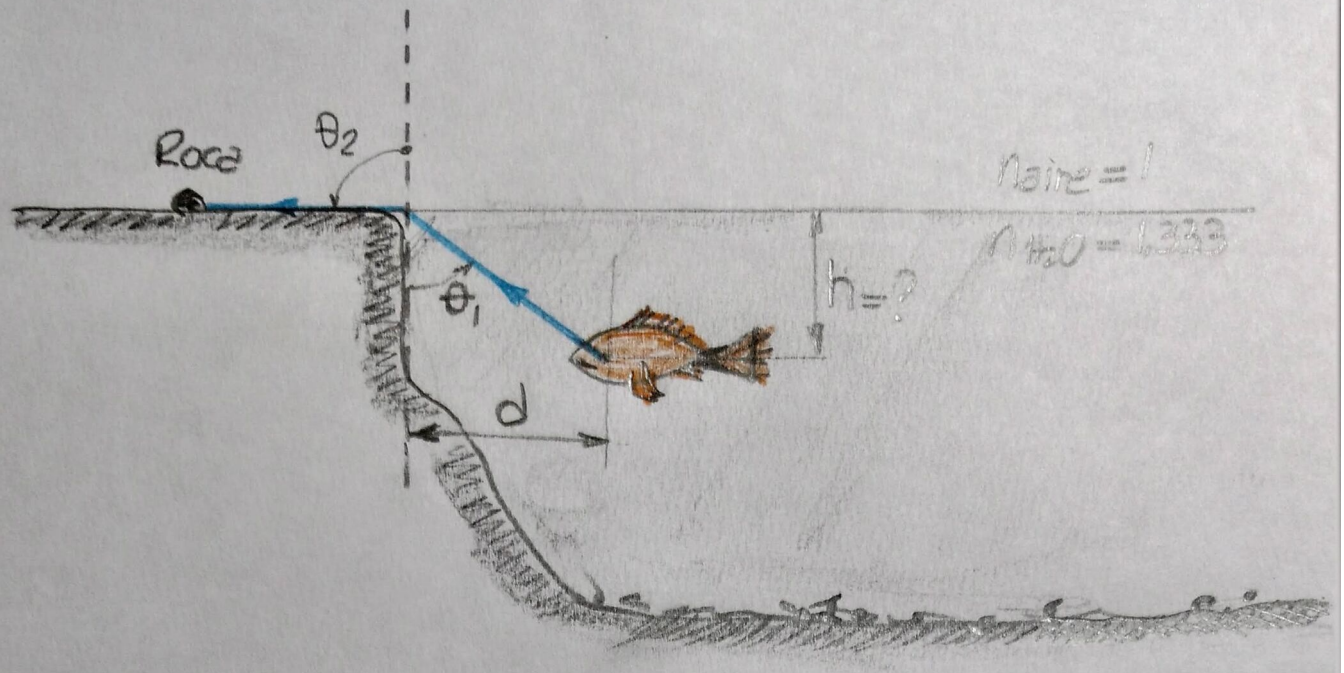
$\theta_3 = 45°$ $\theta_4 = 90°$

$n_V \cdot \operatorname{sen}\theta_3 = n_{H_2O} \cdot \operatorname{sen}$

$$n_V = \frac{n_{H_2O}}{\operatorname{sen} 45°} = 1{,}885$$

$$n_V > 1{,}885$$

34)



$d = 15\,(m)$

$\theta_1 = \theta_{CRITICO}$

$\theta_2 = 90°$

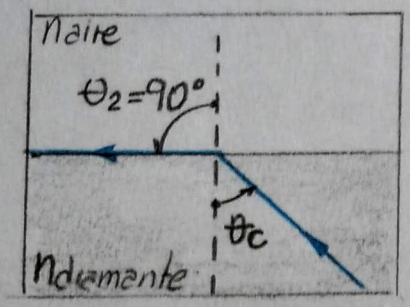
$$n_{H_2O} \cdot \operatorname{sen}\theta_1 = n'_{aire} \operatorname{sen}' 90°$$

$$\theta_1 = \operatorname{arc\,sen} \frac{1}{1,333} = 48,6°$$

$$\operatorname{tg}\theta_1 = \frac{d}{h}$$

$$h = \frac{15\,(m)}{\operatorname{tg} 48,6°} = 13,2\,(m)$$

35)



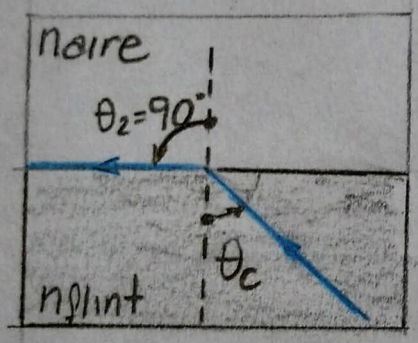
$n_{diamante} = 2{,}419$

$n_{aire} = 1$

$\theta_2 = 90°$

$n_{diam} \cdot \operatorname{sen}\theta_c = n_{aire} \operatorname{sen} 90°$

$\theta_{c\,|\,diamante} = \text{arc sen}\ \frac{1}{2{,}419} = 24{,}4°$



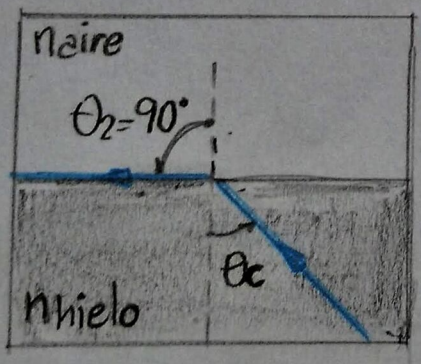
$n_{flint} = 1{,}66$

$n_{aire} = 1$

$\theta_2 = 90°$

$n_{flint} \cdot \operatorname{sen}\theta_c = n_{aire} \operatorname{sen} 90°$

$\theta_{cflint} = \text{arc sen}\ \frac{1}{1{,}66} = 37{,}0°$



$n_{hielo} = 1{,}309$

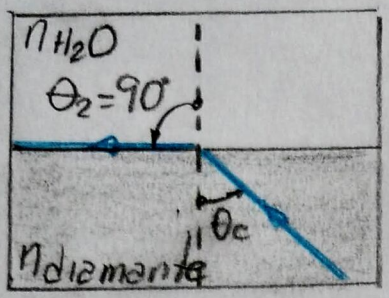
$n_{aire} = 1$

$\theta_2 = 90°$

$n_{hielo} \cdot \operatorname{sen}\theta_c = n_{aire} \operatorname{sen} 90°$

$\theta_{chielo} = \text{arc sen}\ \frac{1}{1{,}309} = 49{,}8°$

36)



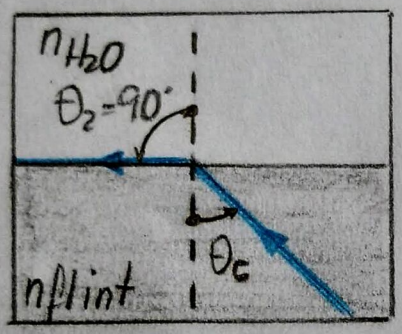
$n_{diamante} = 2{,}419$

$n_{H_2O} = 1{,}333$

$\theta_2 = 90°$

$n_{diam} \cdot \operatorname{sen}\theta_c = n_{H_2O} \cdot \operatorname{sen} 90°$

$\theta c_{\substack{diamante \\ H_2O}} = \text{arc sen } \frac{1{,}333}{2{,}419} = 33{,}4°$



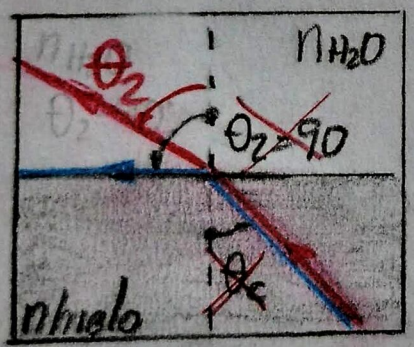
$n_{flint} = 1{,}66$

$n_{H_2O} = 1{,}333$

$\theta_2 = 90°$

$n_{flint} \cdot \operatorname{sen}\theta_c = n_{H_2O} \cdot \operatorname{sen} 90°$

$\theta c_{\substack{flint \\ H_2O}} = \text{arc sen } \frac{1{,}333}{1{,}66} = 53{,}4°$



$n_{hielo} = 1{,}309$

$n_{H_2O} = 1{,}333$

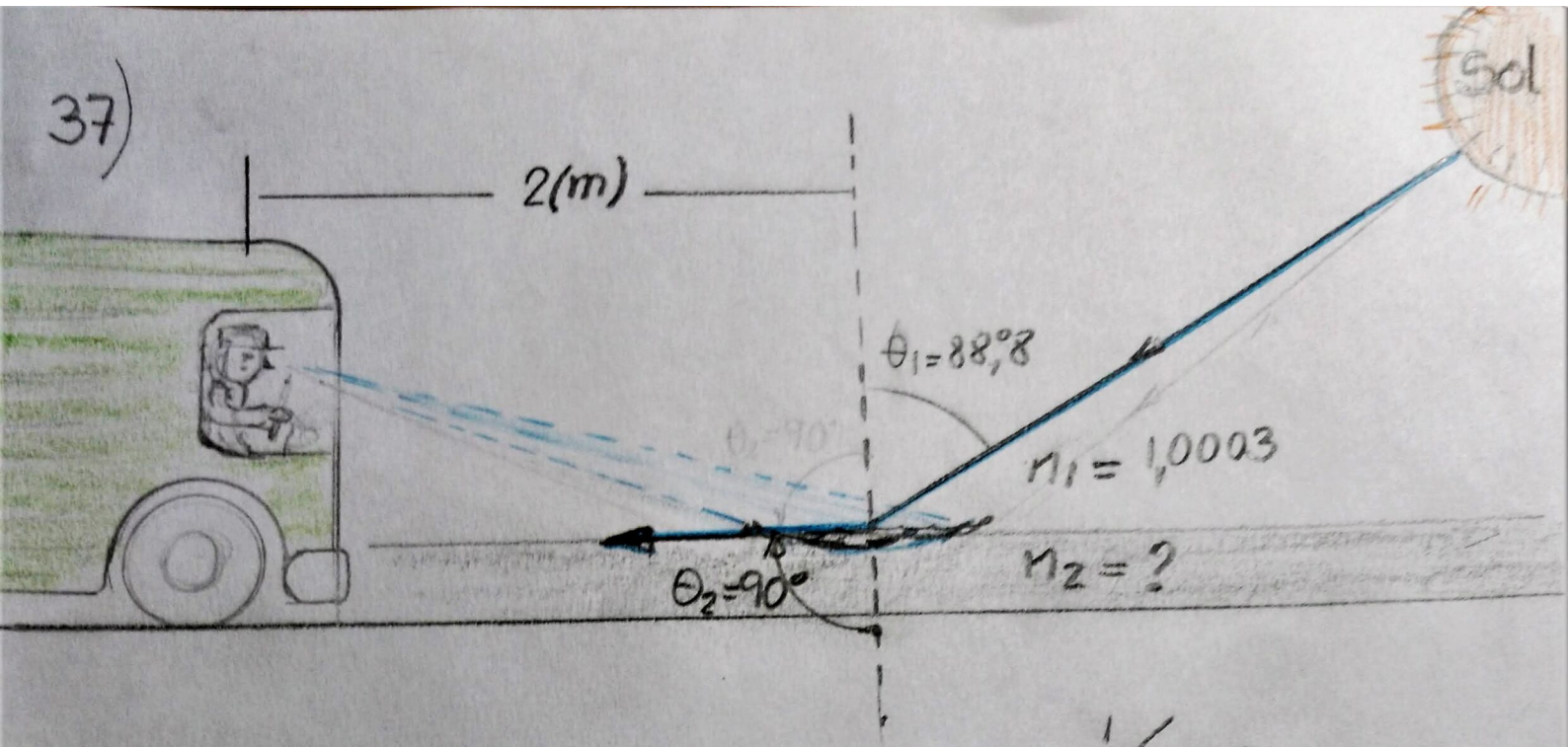
$\theta_2 = 90°$

$n_{hielo} \cdot \operatorname{sen}\theta_c = n_{H_2O} \cdot \operatorname{sen} 90°$

$\theta c_{\substack{hielo \\ H_2O}} = \text{arc sen } \frac{1{,}333}{1{,}309} = \text{no existe}$

el rayo SIEMPRE se REFRACTA

37)

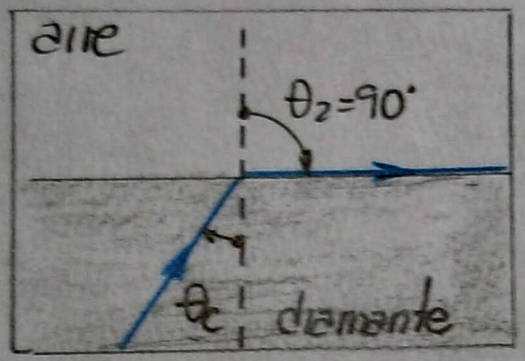


$$n_1 \operatorname{sen} 88,8° = n_2 \operatorname{sen} 90°$$

$$n_2 = 1,0003 \operatorname{sen} 88,8°$$

$$n_2 = 1,00008$$

38)



$n_{diamante} = 2{,}419$

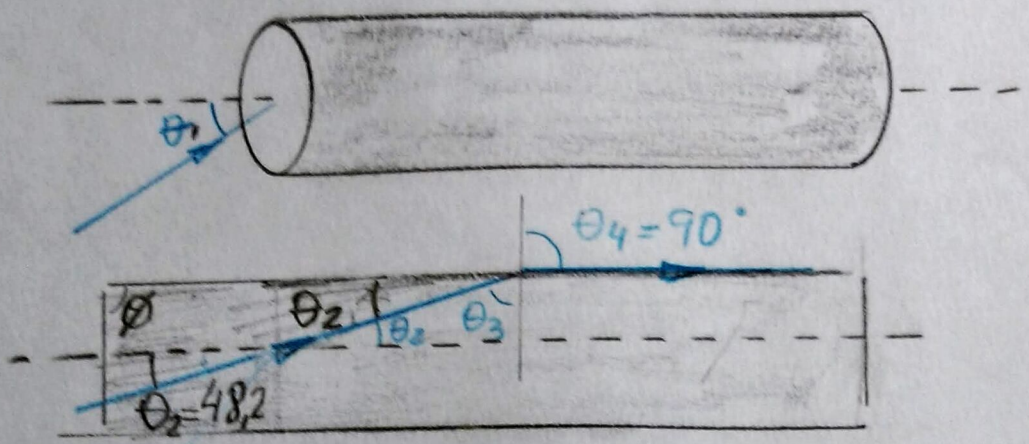
$n_{aire} = 1$

$\theta_2 = 90°$

$n_{diam} \cdot \operatorname{sen}\theta_c = n_{aire} \cdot \operatorname{sen} 90°$

$\theta_c = \operatorname{arc\,sen} \dfrac{1}{2{,}419} = 24{,}4°$

39)



$$n_{aire}^{1} \cancel{sen}^{1}\theta_4 = n_v \, sen\theta_3$$

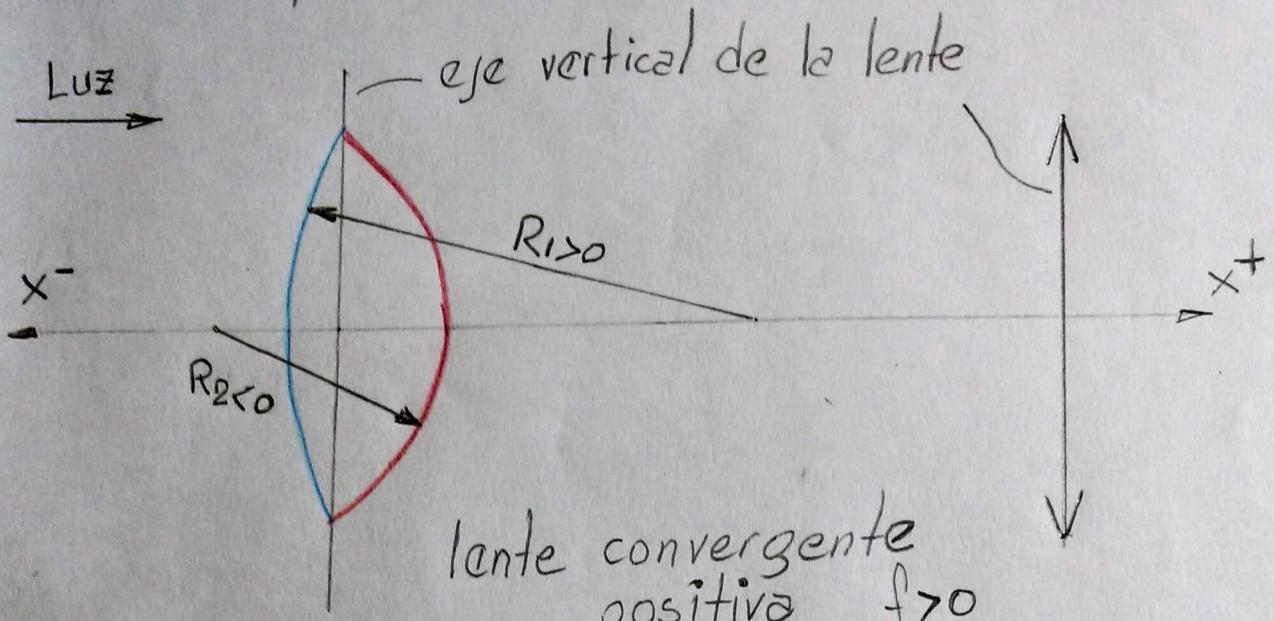
$$\theta_3 = arc\,sen\frac{1}{1,5} = 41,8°$$

$$\theta_2 = \phi - \theta_3 = 90 - 41,8° = 48,2°$$

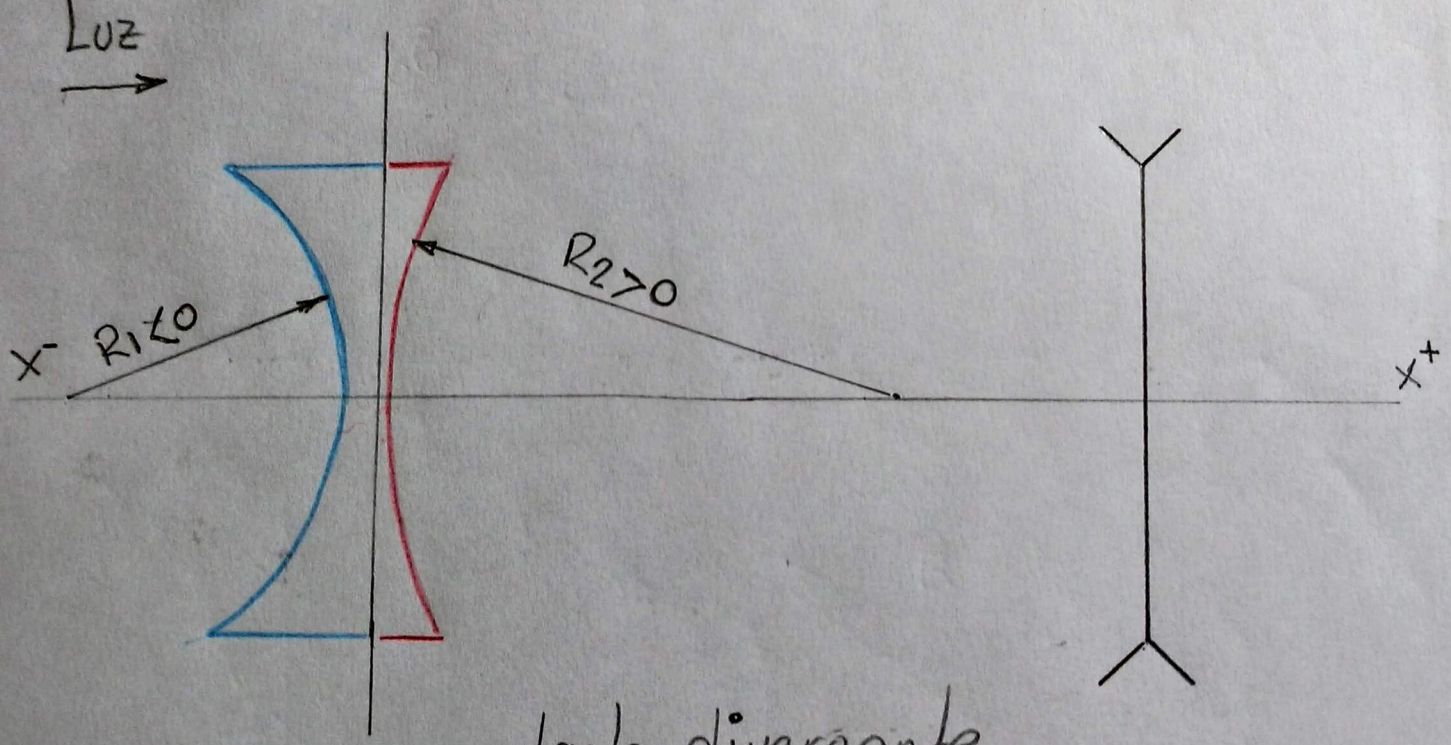
$$n_{aire}^{1} \, sen\theta_1 = n_v . sen\theta_2$$

si calculo θ_1 se verifica que cualquier θ_1 produce reflexión interna

Repaso del teórico



lente convergente
positiva $f>0$
biconvexa



lente divergente
negativa
biconcava

$$\frac{1}{f} = (n_v - 1) \cdot \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{s} + \frac{1}{s'}$$